Quasi-Poisson and Negative Binomial models

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Overdispersion

Overdispersion occurs when the response variable Y_i has greater variability than the model accounts for.

Poisson: $Var(Y_i|\mathbf{x}_i) = \mathbb{E}[Y_i|\mathbf{x}_i] = \mu_i$

For an assumed Poisson model, overdispersion occurs when $Var(Y_i|x_i) > \mu_i$

quasi-Poisson models

$$\log(\mu_i) = \mathbf{x}_i^T \beta$$

$$Var(Y_i|\mathbf{x}_i) = \phi \mu_i \qquad \text{with } \phi > 1 \text{ allowed}$$

$$V(\mu) = \mu \qquad \text{(for Poisson, } \phi = 1\text{)}$$

Estimation:

Poisson regression: $\hat{\beta}$ solves $X^T(Y - \mu) = 0$

update rule: $\hat{\beta}^{(s+1)} = \hat{\beta}^{(s)} + (X^T V X)^{-1} X^T (Y - \hat{\mu}^{(s)})$

$V = \text{diag}(V(\hat{\mu}_i^{(s)}))$

Notice: only requires specifying μ and $V(\mu)$!

$\Rightarrow \hat{\beta}_{QP} = \hat{\beta}_{Poisson}$

Inference: $\hat{\beta} \approx N(\beta, \phi (X^T V X)^{-1})$

For Poisson: $\phi = 1 \Rightarrow \hat{Var}(\hat{\beta}_{Poisson}) = (X^T V X)^{-1}$

For quasi-Poisson: ϕ can be > 1

$\Rightarrow \hat{Var}(\hat{\beta}_{QP}) = \hat{\phi} (X^T V X)^{-1} = \hat{\phi} \hat{Var}(\hat{\beta}_{Poisson})$

Example: articles data

```
glm(art ~ phd + ment + kid5, family = poisson, data = artic
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.276493   0.087294   3.167  0.00154 **
phd          0.008071   0.026358   0.306  0.75944
ment         0.026537   0.002001  13.264  < 2e-16 ***
kid5        -0.099696   0.035248  -2.828  0.00468 **
(Dispersion parameter for poisson family
   taken to be 1)
```

same $\hat{\beta}$'.

$0.0484 = \sqrt{1.8846}\,(0.0352)$

```
glm(art ~ phd + ment + kid5, family = quasipoisson, data =
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *
(Dispersion parameter for quasipoisson family
   taken to be 1.88461)
```

$\hat{\phi}$

Example: articles data

```
glm(art ~ phd + ment + kid5, family = quasipoisson, data =
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *

(Dispersion parameter for quasipoisson family
   taken to be 1.88461)

    Null deviance: 1817.4  on 914  degrees of freedom
Residual deviance: 1661.1  on 911  degrees of freedom
AIC: NA
```

AIC is not defined b/c a quasi-Poisson does not specify a full distribution for Y_i, only mean and variance

t-tests for quasi-Poisson models

$$\widehat{\beta} \approx N(\beta, \phi(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1})$$

$$H_0 : \beta_j = b \qquad Z = \frac{\widehat{\beta}_j - b}{\sqrt{\phi[(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1}]_{jj}}} \approx N(0,1)$$

$$\hat{\phi} = \frac{D(y,\hat{\mu})}{n-p} \quad \text{or} \quad \hat{\phi} = \frac{1}{n-p}\sum_{i=1}\frac{(y_i-\hat{\mu}_i)^2}{V(\hat{\mu}_i)}$$

$$(n-p)\frac{\hat{\phi}}{\phi} \approx \chi^2_{n-p}$$

$$t = \frac{\hat{\beta}_j - b}{\sqrt{\hat{\phi}[(X^TVX)^{-1}]_{jj}}} = Z\cdot\sqrt{\frac{\phi}{\hat{\phi}}} \approx t_{n-p}$$

$$= \frac{Z}{\sqrt{(n-p)\frac{\hat{\phi}}{\phi}/(n-p)}} \leftarrow N(0,1) \qquad \leftarrow \sqrt{\ } \text{ of } \chi^2_d/d$$

F-tests for quasi-Poisson models

The quasi-Poisson does not specify a full distribution.

no likelihood!

But, the deviance still exists.

Nested test:

$$\frac{D_{reduced} - D_{full}}{\phi} \approx \chi^2_{p_{full} - p_{reduced}}$$

$p_{full} - p_{reduced} = q$

$$\frac{\left(\frac{D_r - D_f}{\phi}\right) / q}{\left(\frac{(n-p)\,\hat{\phi}_f}{\phi}\right) / (n - p_{full})} = \frac{(D_r - D_f)/q}{\hat{\phi}_{full}} \approx F_{q,\, n-p_{full}}$$

Negative Binomial models

known constant

(r is not ϕ; for NB, $\phi = 1$)

$$Y \sim NB(\mu, r) \qquad f(y; \mu, r) = \frac{\Gamma(y+r)}{\Gamma(y+1)\Gamma(r)} \left(\frac{\mu}{\mu+r}\right)^y \left(1 - \frac{\mu}{\mu+r}\right)^r$$

$$Y_i | \mathbf{x}_i \sim NB(\mu_i, r) \qquad \log(\mu_i) = \mathbf{x}_i^T \beta$$

not canonical link

$$\text{Var}(Y_i | x_i) = \mu_i + \frac{\mu_i^2}{r}$$

For large r, $\text{Var}(Y_i | x_i) \approx \mu_i$

For small r, $\text{Var}(Y_i | x_i) >> \mu_i$

In practice, r is unknown $\Rightarrow$ not an EDM, need to estimate r

Example: articles data

```
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.233528   0.117585   1.986   0.0470 *
phd          0.008203   0.036069   0.227   0.8201
ment         0.030374   0.003240   9.373   <2e-16 ***
kid5        -0.093256   0.046475  -2.007   0.0448 *

(Dispersion parameter for Negative Binomial(2.1647)
   family taken to be 1)

AIC: 3145.6
```

Inference for negative binomial models

- Can use the usual Wald tests for β (from asymptotic normality)
- LRT also works because a full distribution is specified

Example: articles data

```r
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
m4 <- MASS::glm.nb(art ~ phd + kid5, data = articles)

pchisq(m3$twologlik - m4$twologlik, 1, lower.tail=F)
```

```
## [1] 1.708206e-18
```